ORBIS — Detailed Master Project Checkpoint

React Bits component decisions + complete capability inventory + implementation status + future scope

Purpose of this checkpoint

This document consolidates the ORBIS decisions discussed so far, including the approximately 15 React Bits components that were considered as the intended visual foundation and the expanded computer-assistance functionality discussed for the project. It separates what is genuinely implemented from what is planned, hardware-dependent, or not universally guaranteeable.

Important status rule: “Completed” means the capability exists in the current ORBIS backend/system at a meaningful working level. “Foundation complete” means the underlying architecture exists but the user-facing capability is still to be built. “Future” means it belongs to a later phase. “Limited” means it works only within explicit technical/security boundaries.

1. React Bits — ORBIS Visual Component Set

The following five were explicitly locked as core visual decisions. The remaining items were strong candidates/near-confirmed components discussed for the ORBIS interface and are included here so the design checkpoint does not lose those decisions.

#	Component	Intended ORBIS location	Purpose / functionality	Status
1	Globe	Home / ORBIS Core	Living central 3D globe. Continuously rotates and reacts to actual ORBIS states: listening, reasoning, researching, executing, speaking, approval and completion.	LOCKED / CORE
2	DecryptedText	Startup / Awakening	ORBIT/ORBIS identity reveal when the system opens or awakens.	LOCKED / CORE
3	SplitText	Conversation layer	Animated user↔ORBIS messages and spoken-response presentation. Selective use to keep the interface clean.	LOCKED / CORE
4	LogoLoop	Live Flow / Research	Scrolling ecosystem/research activity visualization driven by real backend events rather than fake progress.	LOCKED / CORE
5	OptionWheel	Primary navigation	Curved interactive navigation signature. Replaces the ordinary three-line hamburger on the main ORBIS experience.	LOCKED / CORE
6	Specter Orb	Ambient ORBIS state / interaction	Candidate for a secondary reactive orb/state visual, if it fits without competing with the main globe.	NEAR-CONFIRMED
7	Agentic Ball	Agent activity / state	Candidate visual for agent activity or an alternate compact state indicator.	NEAR-CONFIRMED
8	TextType	Conversation / status	Candidate typing-style text treatment for concise dynamic ORBIS status or response areas.	NEAR-CONFIRMED
9	Folder	Files / workspace	Candidate visual interaction for workspace files/projects and continuity resources.	NEAR-CONFIRMED
10	ElasticSlider	Controls / settings	Candidate for smooth adjustable settings such as volume, brightness or similar numeric controls in later desktop-control UI.	NEAR-CONFIRMED
11	AnimatedList	Activity / history	Candidate for task history, event lists, research sources or recent workspace activity.	NEAR-CONFIRMED
12	Dock	Contextual tools	Candidate compact access bar for frequently used ORBIS workspaces/tools; should remain contextual rather than cluttering Home.	NEAR-CONFIRMED
13	Stepper	Task / workflow view	Candidate for multi-step task progress where a workflow needs explicit stage representation.	NEAR-CONFIRMED
14	Fluid Glass	Contextual panels	Candidate for premium glassmorphism panels/overlays. Heavy 3D dependencies mean it should be used selectively.	NEAR-CONFIRMED
15	Tilted Card	Cards / contextual information	Candidate for lightweight visual cards such as project/context/source summaries, only where it supports the cinematic design.	NEAR-CONFIRMED

Selection philosophy: these components are not intended to make ORBIS look like a collection of unrelated effects. The Globe remains the dominant identity; DecryptedText handles awakening; SplitText handles conversation; LogoLoop supports real Live Flow/research activity; OptionWheel handles navigation. The additional components should only appear where their interaction is genuinely useful.

2. ORBIS Visual Direction — Locked

Dark • minimal • cinematic • futuristic • restrained JARVIS inspiration. The Home screen is intentionally sparse, with the globe as the focus. Workspaces and panels appear contextually. The design should avoid a generic SaaS dashboard and avoid an overloaded cyberpunk/RGB gaming aesthetic.

3. Complete Capability Inventory — 50 Items

#	Capability	Status	Current / planned behavior
1	Natural-language Q&A;	COMPLETED	Answer ordinary questions through the ORBIS agent/LLM pipeline.
2	Conversational explanations	COMPLETED	Explain technical concepts and adapt responses to the current conversation.
3	Multi-step task planning	COMPLETED	Hierarchical/DAG planning for tasks involving multiple operations.
4	Task execution + verification	COMPLETED	Execute tools, verify results and report success/failure.
5	Dynamic safety/risk assessment	COMPLETED	Risk-aware execution with approval gates for higher-risk operations.
6	Human approval / continuation	COMPLETED	Pending approval can be resumed, approved, denied or invalidated on context switch.
7	Web research	COMPLETED	Controlled web research with source discovery/fetching, ambiguity handling and synthesis.
8	Multi-source research	COMPLETED	Research can gather evidence from multiple sources rather than relying on a single result.
9	Terminal automation	COMPLETED / CONTROLLED	Allowlisted shell-free execution, workspace boundaries, bounded output and secret redaction.
10	File listing	COMPLETED / CONTROLLED	Safe workspace listing with protected entries filtered.
11	File reading	COMPLETED / CONTROLLED	Read supported workspace files while protecting sensitive paths/content.
12	File creation	COMPLETED / CONTROLLED	Create supported text/Markdown files; overwrite can require approval.
13	Document generation	COMPLETED / LIMITED	Markdown/text generation is implemented; richer office formats are future.
14	Workspace observation	COMPLETED / LIMITED	Passive Windows observation and workspace events; some foreground detection depends on environment.
15	Session segmentation	COMPLETED	Groups workspace activity into sessions.
16	Context DNA	COMPLETED	Structured representation of task/session context and relationships.
17	Workspace Knowledge Graph	COMPLETED	NetworkX + SQLite graph representing sessions, files, apps, resources and relationships.
18	Semantic vector memory	COMPLETED	Sentence Transformers + FAISS semantic retrieval.
19	Continuity retrieval	COMPLETED	Retrieve context relevant to questions such as what the user was working on.
20	Intent Continuity Engine (ICE)	COMPLETED	Estimates unfinished work and likely next actions using deterministic/semantic signals.
21	Conversation continuity	COMPLETED	Bounded multi-turn context, reference resolution, topic switching and artifact tracking.
22	Workspace restoration	COMPLETED / LIMITED	Declarative multi-app/window restoration with validation, allowlists and approval.
23	Voice input	COMPLETED	Push-to-talk local speech capture and transcription.
24	Voice output	COMPLETED	Windows SAPI5 local text-to-speech.
25	Live Flow event foundation	COMPLETED	Typed AgentEvents, async EventBus, bounded history, replay and sanitization.
26	Real-time gateway	NEXT	FastAPI WebSocket/SSE bridge for the future desktop UI; Phase 12.8.
27	System volume / mute	FUTURE	Controlled system tool for volume up/down/mute.
28	Brightness control	FUTURE	Controlled display brightness up/down where supported by Windows/device APIs.
29	Battery Saver	FUTURE	Enable/disable supported power-saving settings.
30	Wi-Fi control	FUTURE	Connect/disconnect and supported Wi-Fi operations through controlled system capabilities.
31	Bluetooth control	FUTURE	Supported Bluetooth connection/settings operations.
32	Broader Windows settings	FUTURE	Expose reasonably supported system settings through safe, typed tools rather than unrestricted OS access.
33	Application launching	FUTURE	Open supported applications and desktop resources.
34	Open PDFs	FUTURE / PARTLY AVAILABLE	PDF files can be handled as files today; dedicated desktop opening is a future AppLaunch capability.
35	Multi-application sequencing	FUTURE	Example: open WhatsApp, leave it open, then launch another application/game.
36	YouTube search/play	FUTURE	Controlled browser/media workflow for search/open/play.
37	WhatsApp Web interaction	FUTURE	Supported browser/app interaction where technically possible.
38	Instagram Web interaction	FUTURE	Supported browser/app interaction where technically possible.
39	Messaging	FUTURE / HIGH-SIDE-EFFECT	Sending messages is possible only through an appropriate authorized capability and safety/approval flow.
40	Interactive browser automation	FUTURE	GUI-level browser tasks separate from current web research.
41	DOCX generation/editing	FUTURE	Rich Word document workflows.
42	PPTX generation/editing	FUTURE	Rich presentation generation and modification.
43	PDF generation/workflows	FUTURE	Rich PDF creation/editing/export workflows.
44	XLSX workflows	FUTURE	Spreadsheet creation, editing and analysis.
45	Live document visualization	FUTURE	Show real document/PPT/PDF updates in the UI while a task executes.
46	Live research visualization	FUTURE / FOUNDATION	Show real research milestones, sources and progress through the event stream.

#	Capability	Status	Current / planned behavior
47	Keyboard/RGB control	FUTURE / HARDWARE-DEPENDENT	Keyboard lighting and related controls where hardware/vendor APIs support them.
48	Game download/install	FUTURE / HIGH-RISK	Potential controlled installation workflow such as through Epic Games; requires supported automation and safety approval.
49	Autonomous multi-app workflows	FUTURE	Coordinate multiple applications while maintaining task context and safety.
50	Live Flow disconnect independence	FOUNDATION / DESIGN LOCK	Closing Live Flow will hide the visualization only; it must not stop/cancel the underlying ORBIS task.

4. Status Meaning and Important Boundaries

COMPLETED: the core capability has been implemented and tested at the current intended scope. This does not mean unlimited support for every application or operating-system variation.

COMPLETED / LIMITED: functional but constrained by Windows visibility, supported file formats, application APIs, resource availability or other explicit technical boundaries.

FOUNDATION COMPLETE: the architectural infrastructure exists, but the user-facing capability still needs a gateway/UI or later tool implementation.

FUTURE: intentionally deferred to later ORBIS phases. These should be implemented as controlled tools rather than by bypassing the existing safety architecture.

HIGH-SIDE-EFFECT / HIGH-RISK: actions such as messaging or software installation can cause external changes and therefore require stronger authorization and approval handling.

HARDWARE-DEPENDENT: RGB and vendor-specific hardware controls depend on the actual device, driver and available APIs.

NOT UNIVERSAL: ORBIS is designed to support authorized digital tasks for which a suitable reliable capability exists. It should not be described as having unlimited control over every computer or every application.

5. What ORBIS Explicitly Will Not Do

• Expose private chain-of-thought or hidden model reasoning through Live Flow.

• Expose API keys, passwords, cookies, bearer tokens, .env contents or other protected secrets.

• Use unrestricted keylogging, password capture or surveillance as a shortcut to workspace continuity.

• Pretend that an action happened when the underlying tool did not perform it.

• Use fake Live Flow progress unrelated to actual backend events.

• Bypass existing safety/approval gates simply because a future capability is requested.

• Promise perfect restoration of arbitrary application UI state.

• Promise universal control of hardware or every application.

6. Architecture for the Expanded Vision

Brain → Events → Gateway → UI → Capabilities

The existing ORBIS brain remains the execution authority. The EventBus reports safe operational milestones. Phase 12.8 exposes those events through a gateway. Electron/React later visualizes them. New capabilities such as system control, browser automation and hardware control are introduced as typed tools behind the existing planner, risk, approval and verification boundaries.

Example future task: *“Open WhatsApp, don’t send a message, then start Marvel Rivals.”* ORBIS should interpret this as a multi-step workflow, use supported application-launch capabilities, preserve context between steps, and keep the no-message constraint explicit. It should not infer permission to send a message.

7. Revised ORBIS Roadmap

Phase	Scope	Status
1	Foundation, Tool Contract & Minimal Agent Shell	COMPLETE
2	Passive Workspace Observation & Base Desktop Tools	COMPLETE
3	Session Segmentation & Context DNA	COMPLETE
4	Knowledge Graph & Semantic Vector Store	COMPLETE
5	First Continuity Vertical Slice	COMPLETE
6	Multi-Step Agent Brain & Hierarchical Planner	COMPLETE
7	Web Research & Controlled Terminal Automation	COMPLETE
8	Content & Document Generation	COMPLETE
9	Intent Continuity Engine (ICE)	COMPLETE
10	Multi-App & Window Layout Restoration	COMPLETE
11	Local Voice Interface	COMPLETE
11.5	Voice/Q&A; Hardening + Research + Conversation Continuity	COMPLETE
12.1–12.7	Backend Agent Event Architecture	COMPLETE + COMMITTED
12.8	FastAPI Gateway + Real-Time IPC	NEXT
12.9+	Electron/React ORBIS UI + Live Flow	FUTURE
13	System Control, App Launch & Desktop Capabilities	FUTURE
14	Interactive Web Automation & Office Document Workflows	FUTURE
15	Hardware Integrations, Autonomous Multi-App Workflows & Polish	FUTURE

8. Current Technical Checkpoint

Known-good commit: 23bbd281ad1e47401e8487e21b5daa7f05aed71c — “Complete Phase 12 backend event architecture”.

Verified test state: 397 tests collected, 395 passed, 0 failed, 2 skipped because live Gemini integration tests require GEMINI_API_KEY. Phase 12 event tests: 16/16; remediation security: 14/14; capability stabilization: 13/13.

Phase 12.1–12.7: Event model, EventSanitizer, EventBus, executor/orchestrator integration, tests and documentation are committed.

Immediate next step: Phase 12.8 FastAPI Gateway + Real-Time IPC. No frontend implementation should begin until this gateway layer is reviewed.

9. Final Project Definition

ORBIS is a general-purpose, safety-bounded desktop AI agent whose distinguishing capability is **workspace continuity and context reconstruction**. It combines natural-language/voice interaction, controlled digital tools, research, content creation, workspace memory, intent continuity and restoration. Its future scope expands into system control, application interaction, browser automation, communication, rich documents and hardware integrations where reliable authorized capabilities exist.

The project should be presented as a **real agent platform with continuity intelligence**, not merely a desktop assistant and not merely a session-restoration utility.